BusterNet + VQC Fusion — Quantum-Hybrid Copy-Move Forgery Detection

STAGE 1 | Input & Two-Branch Feature Extraction



BusterNetPreprocess
BGR convert + mean subtract

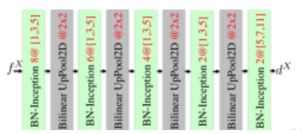
X : 3 x 256 x 256

Mani-Det branch

VGG16
first 4 blocks
(random init)

f_m : 512x16x16

ManiDecoder



d_m : 6 x 256 x 256

Aux head (training only)
Conv2d(6→1) + sigmoid

M_m : *manipulation mask → auxiliary BCE loss*

Simi-Det branch

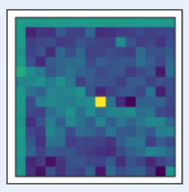
VGG16
first 4 blocks
(ImageNet-pretrained)

f_s : 512x16x16

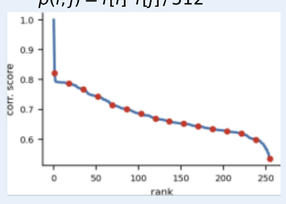
std_norm_along_chs
$\tilde{f}[i] = (f[i] - \mu)/\sigma$

Self-Correlation + Percentile Pooling (paper Eq. 1–6)

$$\rho(i,j) = \tilde{f}[i]^T \tilde{f}[j] \,/\, 512$$



real corr. map, centre patch
S : 16x16x256



sorted scores; red = the K=256 percentile taps

P : 16x16x256

BatchNorm

SimilarityDecoder
BN-Inception + bilinear up x4
(separate weights)

Aux head (training only)
sigmoid

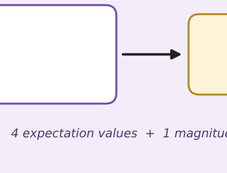
d_s : 6 x 256 x 256 M_s : *similarity mask → auxiliary BCE loss*

STAGE 2 | Fusion & Quantum Decision Layer

Concatenate $d_s \| d_m$

12 x 256 x 256

BnInception(12 → 9) **[KEPT — classical, unchanged]**



9 x 256 x 256

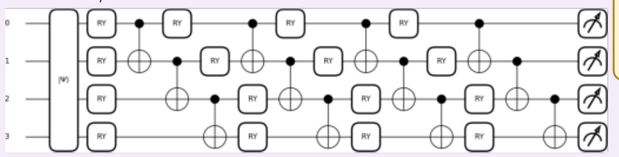
per-pixel reshape [B, 9, H, W] → [B·H·W, 9] · 65,536 circuit calls / image

VQC — Amplitude encoding, 4 qubits · replaces Conv2d(9→3)

BatchNorm1d(9) + ε = 1e-6
(avoids AmplitudeEmbedding's 0/0 on all-zero pixels)

$|\psi_x\rangle = \sum_i x_i |i\rangle$ · L2-normalize, pad 9 → $16 = 2^4$

$\|x\|_2$ side-channel



the actual committed circuit: state prep $|\Psi\rangle$ → 4 x [$R_Y(\theta)$ on 4 qubits + linear CNOT chain] → $\langle Z \rangle$ readout x4

$\langle Z \rangle$ readout
x 4 qubits

concat → 5 values

5 values

4 expectation values + 1 magnitude value

STAGE 3 | Prediction & Output

Reshape back [B·H·W, 5] → [B, 5, H, W]

Conv2d(5 → 3, kernel = 1) [NEW — classical post-layer]
the post-processing layer of the dressed circuit

Softmax over 3 classes, per pixel

Output mask M_c : 3 x 256 x 256
pristine / source copy / target copy



input



target M_c

blue = pristine green = source copy red = target copy

Legend

- Mani-Det branch (manipulation artifacts)
- Simi-Det branch (copy-move similarity)
- Classical fusion / softmax (unchanged)
- Quantum contribution (this project)
- Magnitude side-channel
- Dashed = active during branch training only

Decoder / BN-Inception panels: BusterNet, Wu et al., ECCV 2018, Fig. 3